

BALIKPAPAN SEPINGGAN, IndonesiaWMO: **966330**Lat: **1.2625S**Lon: **116.8947E**Elev: **3**StdP: **101.29**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	23.8	21.6	16.3	26.3	22.0	16.7	26.3	8.4	28.9	7.2	29.4	2.6	0	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	5.3	32.9	27.3	32.2	27.0	31.8	26.9	28.2	31.3	27.7	30.8	27.5	30.5	3.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.0	30.2	27.0	22.6	30.0	26.6	22.1	29.6	90.8	31.5	88.9	31.0	87.2	30.9	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.5	5.7	21.2	34.7	2.4	1.4	19.5	35.7	18.1	36.5	16.8	37.3	15.0	38.3		
(5)			20.3	28.9	2.1	0.6	18.7	29.3	17.5	29.7	16.3	30.0	14.7	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	27.5	27.6	27.8	27.8	28.0	27.6	27.3	27.6	27.9	28.0	27.9	27.7		
	DBStd	1.01	1.07	1.05	0.98	0.99	1.00	0.99	0.96	0.93	0.93	1.03	0.96	0.97		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6474	542	493	552	534	559	528	537	545	538	559	538	550		
	CDD18.3	3432	283	260	293	284	301	278	278	287	288	301	288	292		
	CDH23.3	36689	2791	2635	3058	3016	3386	3031	2934	3094	3247	3407	3103	2986		
	CDH26.7	11487	841	837	989	951	1142	912	791	861	1038	1167	1018	941		
(14) Wind	WSAvg	2.6	2.4	2.7	2.5	2.3	2.4	2.6	3.0	3.3	3.2	2.6	2.2	2.3		
(15) Precipitation	PrecAvg	2705	267	223	255	234	256	240	238	229	177	177	192	238		
	PrecMax	3442	730	378	452	536	550	454	489	759	561	403	382	412		
	PrecMin	1995	57	11	99	117	90	63	36	8	0	20	34	119		
	PrecStd	348	142	96	95	94	105	109	116	164	122	99	98	80		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.2	32.9	32.9	33.1	32.8	31.6	32.1	32.3	33.0	33.1	33.1		
		MCWB	26.5	26.7	26.9	27.6	28.1	27.7	26.9	27.0	27.2	27.4	27.4	27.1		
	2%	DB	31.9	32.1	31.9	31.9	32.0	31.2	30.8	31.0	31.2	31.9	32.0	32.0		
		MCWB	26.5	26.5	26.7	27.1	27.5	27.1	26.7	26.7	26.7	27.0	27.1	26.8		
	5%	DB	31.1	31.3	31.2	31.1	31.2	30.6	30.1	30.2	30.8	31.1	31.2	31.2		
		MCWB	26.4	26.4	26.6	26.9	27.2	26.8	26.4	26.3	26.5	26.7	26.7	26.5		
	10%	DB	30.2	30.5	30.5	30.3	30.5	30.0	29.6	29.7	30.1	30.4	30.6	30.4		
		MCWB	26.1	26.2	26.4	26.6	26.9	26.6	26.2	26.0	26.1	26.4	26.4	26.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	28.0	28.2	28.5	28.7	28.2	27.7	27.6	27.8	28.1	28.2	28.0		
		MCDB	31.0	31.7	31.3	31.6	31.8	31.1	30.1	30.6	31.2	31.5	31.7	31.6		
	2%	WB	27.2	27.2	27.5	27.7	28.0	27.6	27.1	27.0	27.2	27.4	27.5	27.3		
		MCDB	30.4	30.6	30.6	30.7	30.9	30.3	29.7	30.0	30.3	30.7	31.0	30.7		
	5%	WB	26.7	26.8	27.1	27.2	27.6	27.1	26.7	26.6	26.8	27.0	27.1	26.9		
		MCDB	29.9	30.0	30.1	30.1	30.3	29.8	29.3	29.4	29.7	30.2	30.4	30.1		
	10%	WB	26.4	26.4	26.7	26.8	27.1	26.7	26.4	26.2	26.5	26.7	26.7	26.5		
		MCDB	29.3	29.5	29.6	29.6	29.8	29.2	28.8	28.9	29.3	29.7	29.7	29.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.1	6.0	5.6	5.4	5.0	4.8	4.7	4.9	5.3	5.8	6.0		
		MCDBR	7.1	7.1	6.8	6.5	6.4	5.9	5.5	5.3	5.6	6.1	6.7	7.1		
	5% WB	MCWBR	3.1	3.1	3.1	3.1	3.1	3.0	2.8	2.6	2.7	2.8	3.1	3.2		
		MCWBR	6.5	6.6	6.4	6.0	5.9	5.5	5.3	5.1	5.3	5.6	6.4	6.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.405	0.416	0.417	0.410	0.408	0.411	0.419	0.468	0.506	0.538	0.459	0.412		
		taud	2.498	2.458	2.453	2.478	2.501	2.503	2.477	2.310	2.186	2.082	2.330	2.482		
	Ebn at Noon	Ebn at Noon	926	918	908	893	871	855	853	830	818	801	869	915		
		Edh at Noon	113	119	119	112	106	103	107	131	153	171	133	114		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.44	4.87	4.90	4.83	4.67	4.40	4.69	4.96	5.07	4.89	4.58	4.48		
	RadStd	RadStd	0.27	0.46	0.38	0.29	0.21	0.24	0.44	0.45	0.39	0.34	0.20	0.30		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page